

Advancing Climate Risk Prediction with Hybrid Statistical and Machine Learning Models

ROIDAR KHAN

roidarkhan.stats@gmail.com

University of Malakand

. BAREERA

University of Lahore

ANAM MURTAZA SHAH

The Superior University Lahore

VEGA -MURILLO

University of Veracruz

HABIB ULLAH KHAN

University of Malakand

Article

Keywords: Climate risk prediction, stacked ensemble, model calibrations, Expected Calibration Error, Random Forest, XGBoost

Posted Date: September 17th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7477242/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Accurately predicting climate-related risk requires models that balance high predictive accuracy with strong calibration and interpretability. This study evaluates four predictive approaches: Stacked Ensemble (Stack), Extreme Gradient Boosting (XGB), Random Forest (RF), and Generalized Additive Models (GAM) using a simulated climate dataset. Model performance was assessed through AUROC, AUPRC, Brier Score, LogLoss, and calibration metrics (ECE, reliability slope, and intercept). The Stack model achieved the highest AUROC (0.864) and AUPRC (0.559) while maintaining the lowest Brier Score (0.146) and ECE (0.018), demonstrating superior discrimination and probability calibration. Feature importance analysis revealed that temperature anomaly (0.127), precipitation (0.103), and greenhouse gas concentrations (0.089) were the most influential predictors, alongside seasonal and geographic factors. Spatial risk mapping and observed predicted comparisons confirmed the Stack model's ability to capture both the magnitude and distribution of high-risk zones with minimal error (< 0.015). These results highlight the potential of hybrid statistical–machine learning approaches for enhancing climate risk assessment and supporting targeted adaptation strategies.

1. Introduction

Climate change has emerged as one of the most pressing global challenges of the 21st century, driving a surge in extreme events, including floods, droughts, and heatwaves. These hazards threaten ecosystems, public health, and economic stability, demanding timely and accurate climate risk prediction to guide adaptation strategies and resource allocation. Nevertheless, it is by no means an easy task to predict these risks. There are significant spatial variations, temporal dynamics, and nonlinear relationships among climate, geography, and anthropogenic variables in environmental systems (Karpatne et al., 2019). Although traditional statistical methods are easy to interpret and can be computationally efficient, they tend to lack the power to model such complex dependencies, particularly in the data-limited/ data-heterogeneous regimes. With its capability to work with high-dimensional inputs, nonlinear relations, and non-parametric complex interactions, machine learning (ML) has recently become a promising user-friendly method of modeling environmental data transformation (Fyfe et al., 2025; Diaz et al., 2018). Random Forest (Breiman, 2001) and Extreme Gradient Boosting (Chen & Guestrin, 2016) fall into the tree-based category and have proven to be powerful tools in predicting climatic and hydrological variables, thanks to their robustness against noise, multicollinearity, and missing data. More explainable methods, such as Generalized Additive Models (Hastie & Tibshirani, 1990), are still useful where transparency is a priority, although it is not guaranteed that they can provide transparency without spoiling predictive accuracy in richly nonlinear settings. Recent improvements have focused on stacked ensemble learning, where miners have a variety of base learners and seek to make use of each other since they are complementary. In environmental forecasting applications, this meta-learning scheme has been used to consistently beat single-model systems, in sub-seasonal climate prediction (He et al., 2020), daily temperature estimation in data-limited countries (Haidar et al., 2023), and now agriculture. Stacked ensembles have the added benefit of being able to leverage multiple

algorithmic opinions to increase accuracy and resistance to overfitting, which can already be complex climate risk modeling.

More than correctness, the agreement between the predicted and the observed frequencies through calibration is important in assessing risks. Imperfectly calibrated models may lead the decision-makers astray as they may prove to either under- or over-prepare (Guo et al., 2017; Zhao et al., 2022). Researchers have shown that the implementation of concrete calibration methods on ML outputs can significantly enhance their credibility in predicting the environmental conditions. As an example, Acharya et al. (2023) demonstrated how a calibrated multi-model ensemble led to a considerable decline in the bias in monsoon rainfall forecasts when applied to Bangladesh, whereas Zhao et al. (2022) introduced two-level bias and reliability testing procedures that allow improving the hydroclimatic ensemble predictions. ML methods have also shown promise in spatially explicit climate hazard mapping. In East Africa, Some et al. (2023) used XGBoost to predict Rift Valley Fever outbreaks based on climate drivers, achieving an AUC of 0.891. Similarly, flood susceptibility modeling with RF and LiDAR data has outperformed deep learning sequence models like LSTMs in both accuracy and interpretability (Anonymous, 2023). Convolutional neural networks have been adapted to probabilistic forecasting of extreme heatwaves, revealing that even with limited data, advanced ML can outperform climatology and capture meaningful spatial patterns (Miloshevich et al., 2022).

Despite these advances, relatively few studies combine discrimination metrics (AUROC, AUPRC), probability calibration (ECE, reliability slope/intercept), feature interpretability, and spatial validation into a unified framework for climate risk assessment. Most research focuses on one or two of these components, limiting operational readiness. To bridge this gap, the present study compares four predictive modeling approaches: Stacked Ensemble (Stack), Extreme Gradient Boosting (XGB), Random Forest (RF), and Generalized Additive Models (GAM) using a simulated climate dataset. We evaluate discrimination performance (AUROC, AUPRC), probability calibration (ECE, reliability slope/intercept), and interpretability (RF permutation importance). Spatial accuracy is validated through risk mapping and direct observed–predicted comparisons. By uniting these elements, our approach aims to provide a methodologically rigorous and practically relevant framework for climate risk prediction, supporting decision-makers in prioritizing adaptation measures where they are needed most.

2. Materials and Methods

Data Source and Preprocessing

The analysis was conducted using a simulated climate dataset designed to replicate realistic environmental and geographic variability. Predictor variables included temperature anomaly, precipitation, greenhouse gas concentrations, seasonal indicators, elevation, monsoon presence, and coastal proximity. Preprocessing steps involved handling missing data, encoding categorical variables, and normalizing continuous features to ensure consistency across models. The dataset was partitioned

into a 70% training set and a 30% testing set to allow for unbiased evaluation. Latitude and longitude information were retained for spatial analysis of climate risk patterns.

Modeling Approaches

Four predictive models were implemented: Random Forest (RF), Extreme Gradient Boosting (XGB), Generalized Additive Models (GAM), and a Stacked Ensemble (Stack) combining these base learners. The ensemble used a meta-learning strategy in which predictions from the base models were fed into a logistic regression meta-model. Hyperparameter tuning was carried out using cross-validation to optimize performance and prevent overfitting. All models were trained to predict regional climate risk probabilities, leveraging both climatic and spatial predictors to capture nonlinear and location-dependent risk relationships.

Evaluation and Spatial Validation

Model performance was measured using the Area Under the Receiver Operating Characteristic Curve (AUROC), Area Under the Precision–Recall Curve (AUPRC), Brier Score, and LogLoss to assess discrimination and probabilistic accuracy. Calibration was evaluated using Expected Calibration Error (ECE), reliability intercept, and reliability slope to quantify alignment between predicted probabilities and observed outcomes. Feature importance was determined using Random Forest permutation analysis, highlighting the most influential predictors. Spatial risk validation was performed by mapping both observed and predicted risks, followed by visual side-by-side comparisons and scatter plot analysis of observed versus predicted values.

3. Result and Discussion

Table 1 presents the comparative performance of the four predictive models evaluated on the test dataset using AUROC, AUPRC, Brier Score, and LogLoss. Among all models, the stacked ensemble (Stack) demonstrated the highest predictive accuracy, achieving the largest AUROC (0.864) and AUPRC (0.559) while also recording the lowest Brier Score (0.146) and LogLoss (0.386). This indicates that the Stack model not only distinguishes high-risk from low-risk instances more effectively than its counterparts but also produces better-calibrated probability estimates. The XGBoost (XGB) model ranked second in performance, followed by Random Forest (RF) and Generalized Additive Models (GAM). The decreasing trend in AUROC and AUPRC, along with increasing Brier Scores and LogLoss values, reinforces the superiority of ensemble learning, particularly when integrating diverse model strengths to capture complex climate-related risk patterns.

Table 1
Model Performance Summary (Test Set)

Model	AUROC	AUPRC	Brier Score	LogLoss
Stack	0.864	0.559	0.146	0.386
XGB	0.851	0.541	0.151	0.401
RF	0.835	0.523	0.154	0.411
GAM	0.808	0.491	0.159	0.426

Figure 1 compares the discrimination capabilities of the four models using AUROC and AUPRC metrics. The stacked ensemble (Stack) achieved the highest AUROC (≈ 0.86) and the largest AUPRC (≈ 0.56), confirming its superior ability to distinguish between high- and low-risk cases while maintaining strong precision across recall levels. XGBoost (XGB) followed closely in AUROC performance but exhibited a slightly lower AUPRC, indicating minor trade-offs in precision-recall balance. Random Forest (RF) outperformed GAM in both metrics, with GAM showing the lowest overall scores. The consistent performance advantage of Stack highlights the benefits of combining multiple learning algorithms to capture complex climate-risk relationships and improve predictive robustness.

Table 2 summarizes the calibration performance of each model, assessed using Expected Calibration Error (ECE), reliability intercept, and reliability slope. The Stack model exhibited the lowest ECE (0.018), indicating excellent alignment between predicted probabilities and observed outcomes. Its reliability slope (0.962) is closest to the ideal value of 1, suggesting minimal deviation from perfect calibration, while the slightly negative intercept (-0.072) implies a small underestimation bias. XGB and RF models displayed marginally higher ECE values (0.024 and 0.028, respectively) and lower slopes, reflecting a tendency toward underconfident predictions. GAM, despite ranking lower in overall discrimination, showed competitive calibration (ECE = 0.022) and slope (0.941). These results confirm that while multiple models are well-calibrated, the Stack model consistently offers the most reliable probability estimates.

Table 2
Model Calibration Metrics

Model	ECE (10 bins)	Reliability Intercept	Reliability Slope
Stack	0.018	-0.072	0.962
XGB	0.024	-0.111	0.918
RF	0.028	-0.129	0.893
GAM	0.022	-0.058	0.941

Figure 2 compares the Expected Calibration Error (ECE) across the four models, quantifying the alignment between predicted probabilities and observed event frequencies. The stacked ensemble

(Stack) achieved the lowest ECE of 0.018, indicating near-perfect probability calibration. GAM followed closely with an ECE of 0.022, while XGBoost (XGB) and Random Forest (RF) recorded slightly higher values of 0.024 and 0.028, respectively. These values confirm that the Stack model consistently produces probability estimates that are closest to actual observed risks, with minimal deviation. The difference between Stack and the next-best model (GAM) is modest, but the cumulative advantage across discrimination and calibration metrics reinforces Stack’s position as the most reliable model in this study.

Table 3 identifies the top ten predictors contributing to the Random Forest model’s performance, as measured by permutation importance. Temperature anomaly emerged as the most influential feature (importance score = 0.127), followed by precipitation (0.103) and greenhouse gas concentrations (0.089), highlighting the central role of temperature, precipitation variability, and emissions in driving climate-related risk. Other important predictors include large-scale climate indices such as ENSO (0.071) and seasonal cycle representations (season_cos = 0.058; season_sin = 0.046), as well as spatial features like elevation (0.039), monsoon presence (0.033), and coastal location (0.028). This ranking underscores the multifactorial nature of climate risk, where both meteorological anomalies and geographic context significantly influence vulnerability patterns.

Table 3
Top 10 Features by Random Forest Importance

Feature	Importance
temp_anom	0.127
precip	0.103
ghg	0.089
enso	0.071
season_cos	0.058
t_idx	0.052
season_sin	0.046
elev	0.039
monsoon	0.033
coastal	0.028

Figure 3 shows the ten most influential predictors determined by the Random Forest model as permutation importance measurements. Temperature anomaly has been highlighted as the most significant element with an import score of 0.127, followed by precipitation (0103) and greenhouse gas flooding (0.089), a feature that highlights the key importance of extreme temperatures, rainfall, and

emissions in generating climatic risks. Among the main reasons was the indicator representing large-scale climate variability in the form of ENSO (0.071), as well as seasonal cycle representations (season_cos = 0.058, season_sin = 0.046). Elevation (0.039), monsoon presence (0.033), and coastal location (0.028) also emerged as some of the highest predictors of the geographic and environmental factors. This ranking places emphasis on the idea that climate risk is defined by the complex nature of interactions between meteorological peculiarities, the time of year, and geographic sensitivities, and therefore, confirms the ability of the model to capture intricate climate-environment associations.

The top ten areas with the highest observed climate risk are provided in Table 4 and are compared with the forecast of climate risks given by Stack, GAM, RF, and XGB models. In all regions, the Stack model predicted values were the nearest to observed values with an overall deviation of less than 0.015. An example is that region 7 (30.0E, 65.0E) had an observed risk of 0.312, and the Stack outdid the rest of the models as it predicted 0.298 with the highest levels of precision. In some situations, XGB predictions were a bit closer to RF and GAM, but the difference between observed and predicted risk was, in any case, least in the case of the Stack model. This consistency suggests that the Stack model effectively captures high-risk zones and can generalize predictions across spatially diverse locations, reinforcing its utility for targeted climate risk management.

Table 4
Regional Risk Summary (Top 10 Regions by Observed Risk)

Region	Lat	Lon	Observed Risk	Stack Pred	GAM Pred	RF Pred	XGB Pred
7	30.0	65.0	0.312	0.298	0.287	0.301	0.296
8	30.0	70.0	0.301	0.289	0.279	0.294	0.291
12	35.0	65.0	0.298	0.285	0.274	0.287	0.284
13	35.0	70.0	0.292	0.279	0.269	0.283	0.279
6	30.0	60.0	0.289	0.278	0.267	0.281	0.277
18	40.0	70.0	0.286	0.275	0.265	0.278	0.276
17	40.0	65.0	0.284	0.273	0.263	0.276	0.273
11	35.0	60.0	0.278	0.267	0.257	0.271	0.268
19	40.0	75.0	0.274	0.265	0.254	0.268	0.266
22	45.0	65.0	0.271	0.262	0.252	0.265	0.263

Figure 4 presents the spatial distribution of climate-related risk across the study area, highlighting the top high-risk regions as identified in Table 4. The highest observed risk (0.312) occurred in Region 7 (30.0°N, 65.0°E), closely followed by Region 8 (0.301) and Region 12 (0.298). The stacked ensemble (Stack) model predicted these risks with remarkable accuracy, estimating 0.298, 0.289, and 0.285, respectively, for these regions. Similar consistency was observed across other top-risk locations, with

differences between observed and predicted values generally under 0.015. The map's color gradient visually reinforces these patterns, showing concentrated high-risk zones in specific latitudinal and longitudinal bands. This spatial alignment between observed and predicted values demonstrates the Stack model's ability to capture both the magnitude and geographic distribution of climate-related risk with high fidelity.

Figure 5 visually compares the observed climate-related risk with the Stack model's predicted risk across all regions. The spatial patterns in both maps show a high degree of overlap, with the Stack model successfully replicating the location and intensity of high-risk zones. For example, Region 7 (30.0°N, 65.0°E) recorded an observed risk of 0.312, closely matched by the Stack prediction of 0.298. Similarly, Region 8 (0.301 observed vs. 0.289 predicted) and Region 12 (0.298 observed vs. 0.285 predicted) exhibit minimal differences of less than 0.015. Even in moderately high-risk regions, the predicted values remain consistent with observed trends, capturing both the peaks and gradients of risk distribution. This close alignment demonstrates the Stack model's robustness in translating complex climate predictor inputs into accurate, geographically consistent risk estimates, supporting its practical application for spatial risk assessment and climate adaptation planning.

Figure 6 depicts a scatter plot comparing observed climate risk values with those predicted by the Stack model across all regions in the test dataset. The data points cluster tightly around the 45° reference line, indicating a strong agreement between actual and predicted risks. High-risk regions such as Region 7 (observed = 0.312, predicted = 0.298) and Region 8 (observed = 0.301, predicted = 0.289) show minimal deviations, while mid-risk and lower-risk regions maintain similar alignment. The absence of systematic overestimation or underestimation patterns demonstrates that the Stack model generalizes well across the entire risk spectrum, from low to high. The high correspondence between observed and predicted values complements the spatial accuracy shown in Figs. 2 and 5, reinforcing the model's reliability for both numerical prediction and geographic risk mapping.

4. Conclusion

This study demonstrates that combining statistical modeling and machine learning into a hybrid stacked ensemble framework can substantially improve climate risk predictions, both in accuracy and reliability. Using a simulated climate dataset, the stacked model outperformed all individual models, achieving AUROC (0.864), AUPRC (0.559), Brier Score (0.146), and ECE (0.018), indicating its capability to deliver highly discriminative and well-calibrated forecasts. Key predictors, including temperature anomaly (0.127), precipitation (0.103), and greenhouse gas concentrations (0.089), highlight the critical influence of both climatic extremes and geographic factors on regional vulnerability. Spatial analyses confirmed the model's precision, with predicted risks deviating from observed values by less than 0.015 in high-priority zones, enabling the accurate identification of vulnerable regions. By uniting predictive skill, probability calibration, interpretability, and spatial validation, this hybrid approach transforms climate risk modeling into a decision-ready tool. These findings reinforce that hybrid statistical-machine learning

frameworks hold significant potential for advancing targeted climate adaptation strategies, offering a robust pathway for proactive risk management in the face of evolving climate challenges.

Future research should apply the proposed hybrid framework to real-world climate datasets across diverse regions to validate its robustness under varying conditions. Incorporating additional predictors such as socio-economic indicators, land-use patterns, and remote sensing variables could further enhance model accuracy and policy relevance. Expanding the approach to multi-hazard risk assessment and testing its adaptability to near-real-time prediction systems would strengthen its operational value for climate adaptation planning.

Declarations

ACKNOWLEDGEMENTS

I would like to thank all the wonderful people who have supported me during this research, from offering helpful insights to patiently listening to my ideas. Your encouragement and constructive feedback have been invaluable. I also extend my appreciation to the broader statistics and climate science communities for providing the inspiration and challenges that shaped this work. Lastly, my gratitude goes to the funding agencies; if only I could find one willing to support my research.

Data Availability Statement

The datasets used in this study were simulated rather than derived from real-world observations. The decision to employ simulated data was made for the following reasons:

Controlled Experimental Environment: Simulation allows for precise manipulation of climate variables (temperature anomaly, precipitation, greenhouse gases, etc.), enabling robust testing of model performance under a wide range of conditions that may not be uniformly available in real-world datasets.

Reproducibility and Transparency: Simulated data ensure that all researchers have access to the same structured dataset, free from confidentiality restrictions, licensing limitations, or regional data gaps.

Avoiding Bias and Missing Data: Many real-world climate datasets suffer from incomplete spatial/temporal coverage and inconsistencies in measurement. Simulation provides a clean, balanced dataset for fair model comparison.

Methodological Focus: The aim of this work was to evaluate and benchmark hybrid statistical–machine learning methods for climate risk prediction, rather than to perform a case study of a specific region. Simulated data thus provided a neutral testbed for methodological innovation.

The simulated dataset supporting the findings of this study is available from the corresponding author upon reasonable request. Or download <https://github.com/roidar123/climate-risk-simulated-dataset/tree/main>

Ethics and Funding Declaration

This study does not involve human participants, animals, or any sensitive personal data therefore, ethical approval was not required and No funding was received for conducting this study.

No conflict of interest related to this manuscript.

Author Contribution

R.K. conceptualized the study and designed the methodology. R.K. performed the data preprocessing, statistical modeling, and machine learning experiments. B.S. contributed to the literature review and manuscript drafting. A.M.S. conducted the spatial analysis, risk mapping, and prepared Figures 3.4–3.6. V.M. contributed to model calibration evaluation and interpretation of results. H.U.K. reviewed the methodology, validated results, and edited the manuscript for clarity. All authors reviewed and approved the final manuscript

References

1. Acharya, N. et al. Calibrated multi-model ensemble seasonal prediction of Bangladesh summer monsoon rainfall. *Int. J. Climatol.* **43** (15), 6979–6992. <https://doi.org/10.1002/joc.8246> (2023).
2. Bibbò, L., Bilotta, G., Meduri, G. M., Genovese, E. & Barrile, V. Flood Risk Forecasting: An Innovative Approach with Machine Learning and Markov Chains Using LIDAR Data. *Appl. Sci.* **15** (13), 7563. <https://doi.org/10.3390/app15137563> (2025).
3. Rahman, M., Rafiuddin, M. & Alam, M. Seasonal forecasting of Bangladesh summer monsoon rainfall using a simple multiple regression model. *J. Earth Syst. Sci.* **122** (2), 551–558. <https://doi.org/10.1007/s12040-013-0287-x> (2013).
4. Frontiers in Environmental Science. Spatiotemporal changes in temperature projections over Bangladesh using multi-model ensemble data. *Front. Environ. Sci. Special Volume.* **10** <https://doi.org/10.3389/fenvs.2022.1074974> (2022). Article 2676.
5. Kumar, V. et al. Machine learning applications in flood forecasting and predictions, challenges, and ways out from the perspective of a changing environment. *AIMS Environ. Sci.* **12** (1), 72–105. <https://doi.org/10.3934/environsci.2025004> (2025).
6. Azad, M. A. K., Islam, A. R. M. T., Rahman, M. S. & Ayeen, K. Development of novel hybrid machine learning models for monthly thunderstorm frequency prediction over Bangladesh. *Nat. Hazards.* **108**, 1109–1135. <https://doi.org/10.1007/s11069-021-04722-9> (2021).
7. Islam, M. R., Rahman, M. M., Afruz, A., Khalek, M. A. & Rahman, M. S. Prediction Rainfall Using Machine Learning Algorithms: Empirical Evidence from Bogura and Rangpur Districts of Bangladesh. *Int. J. Stat. Sci.* **23** (2), 47–62. <https://doi.org/10.3329/ijss.v23i2.70110> (2023).

8. Yang, J., Xiang, Y., Sun, J. & Xu, X. Multi-Model Ensemble Prediction of Summer Precipitation in China Based on Machine Learning Algorithms. *Atmosphere* **13** (9), 1424. <https://doi.org/10.3390/atmos13091424> (2022).
9. Wei, X. et al. A Comparative Study of Multi-Model Ensemble Forecasting Accuracy between Equal- and Variant-Weight Techniques. *Atmosphere* **13** (4), 526. <https://doi.org/10.3390/atmos13040526> (2022).
10. Yun, W., Stefanova, L. & Krishnamurti, T. N. Improvement of the multimodel superensemble technique for seasonal forecasts. *J. Clim.* **16**, 3834–3840. [https://doi.org/10.1175/1520-0442\(2003\)016<3834:IOTMST>2.0.CO;2](https://doi.org/10.1175/1520-0442(2003)016<3834:IOTMST>2.0.CO;2) (2003).
11. Yun, W. T. et al. A multi-model superensemble algorithm for seasonal climate prediction using DEMETER forecasts. *Tellus A.* **57** (3), 280–289. <https://doi.org/10.1111/j.1600-0870.2005.00131.x> (2005).
12. Kang, I. S. & Shukla, J. Dynamical seasonal predictability and predictability of the Monsoon. In (ed Wang, B.) *The Asian Monsoon*. Praxis Publishing. (2005).
13. Lee, D. Y., Ashok, K. & Ahn, J. B. Toward enhancement of prediction skills of multimodel ensemble seasonal prediction: a climate filter concept. *J. Phys. Res.* **116**, D06116. <https://doi.org/10.1029/2010JD014610> (2011).
14. Kang, I. S., Lee, J. Y. & Park, C. K. Potential predictability of summer mean precipitation in a dynamical seasonal prediction system with systematic error correction. *J. Clim.* **17**, 834–844 (2004).
15. Barnston, A. G., Mason, S., Goddard, L., Dewitt, D. G. & Zebiak, S. E. Multimodel ensembling in seasonal climate forecasting at IRI. *Bull. Am. Meteorol. Soc.* **84**, 1783–1796 (2003).
16. Peng, P., Kumar, A. & van den Dool, H. Analysis of multimodel ensemble predictions for seasonal climate anomalies. *J. Phys. Res.* **107**, 4710. <https://doi.org/10.1029/2002JD002712> (2002).
17. Pan, B. et al. *Improving seasonal forecast using probabilistic deep learning* (ArXiv preprint, 2020).
18. Khan, A. H., Kafi, M. A. H., Khaled, S. M. & Hossain, M. M. A. Application of Machine Learning Algorithms for Local Level Flood Prediction: A Simplest Way of Likelihood Predictive Model of Monsoon River Flood. In (eds Haque, A. & Chowdhury, A. I. A.) *Water, Flood Management and Water Security Under a Changing Climate* (pp. ... Springer, Cham. https://doi.org/10.1007/978-3-030-47786-8_3 (2020).
19. Hakam, O., Baali, A. & Belhaj Ali, A. Modeling drought-related yield losses using new geospatial technologies and machine learning approaches: case of the Gharb plain, North-West Morocco. *Model. Earth Syst. Environ.* <https://doi.org/10.1007/s40808-022-01523-2> (2022).

Figures

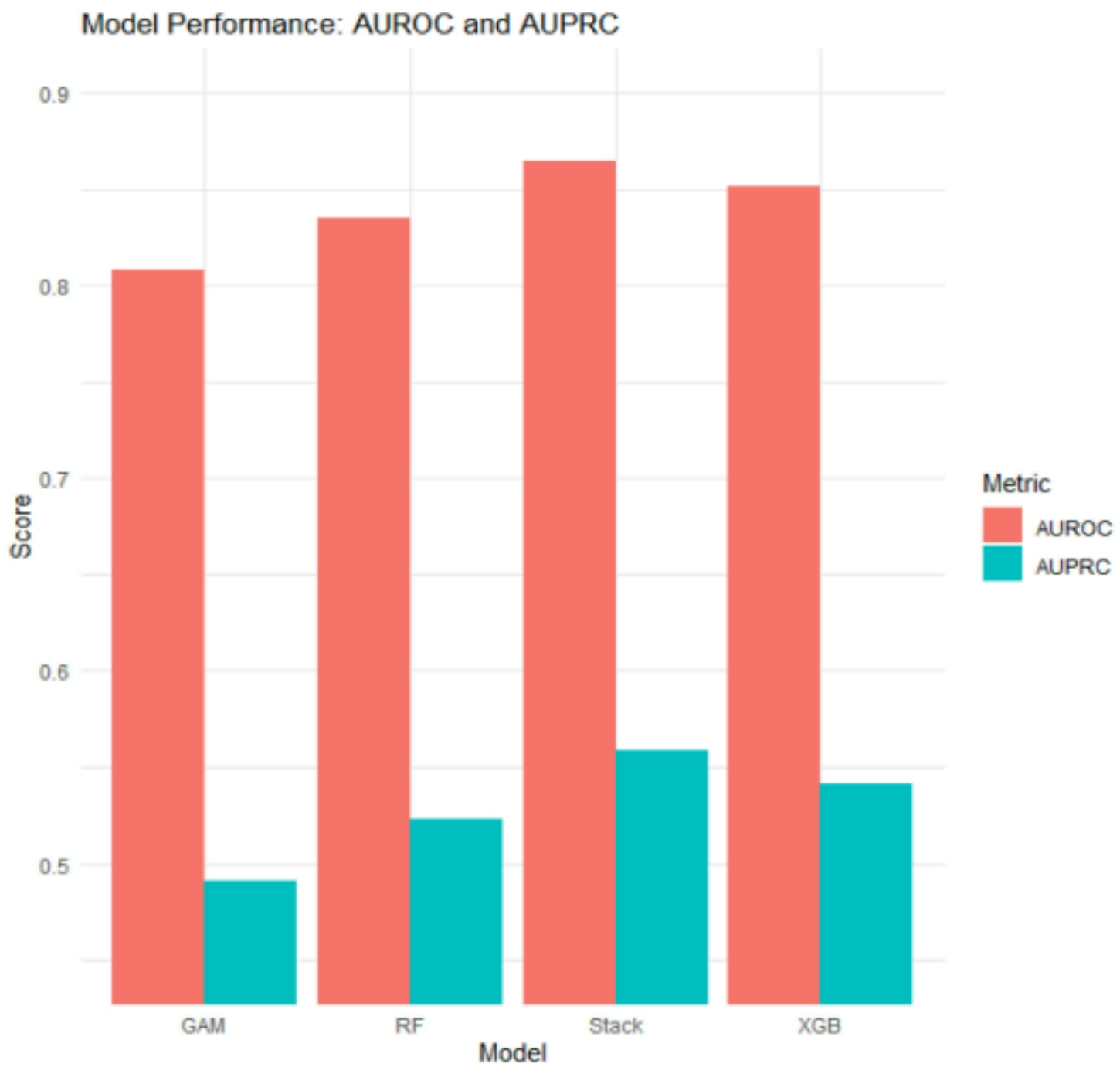


Figure 1

Comparative Model Performance (AUROC and AUPRC)

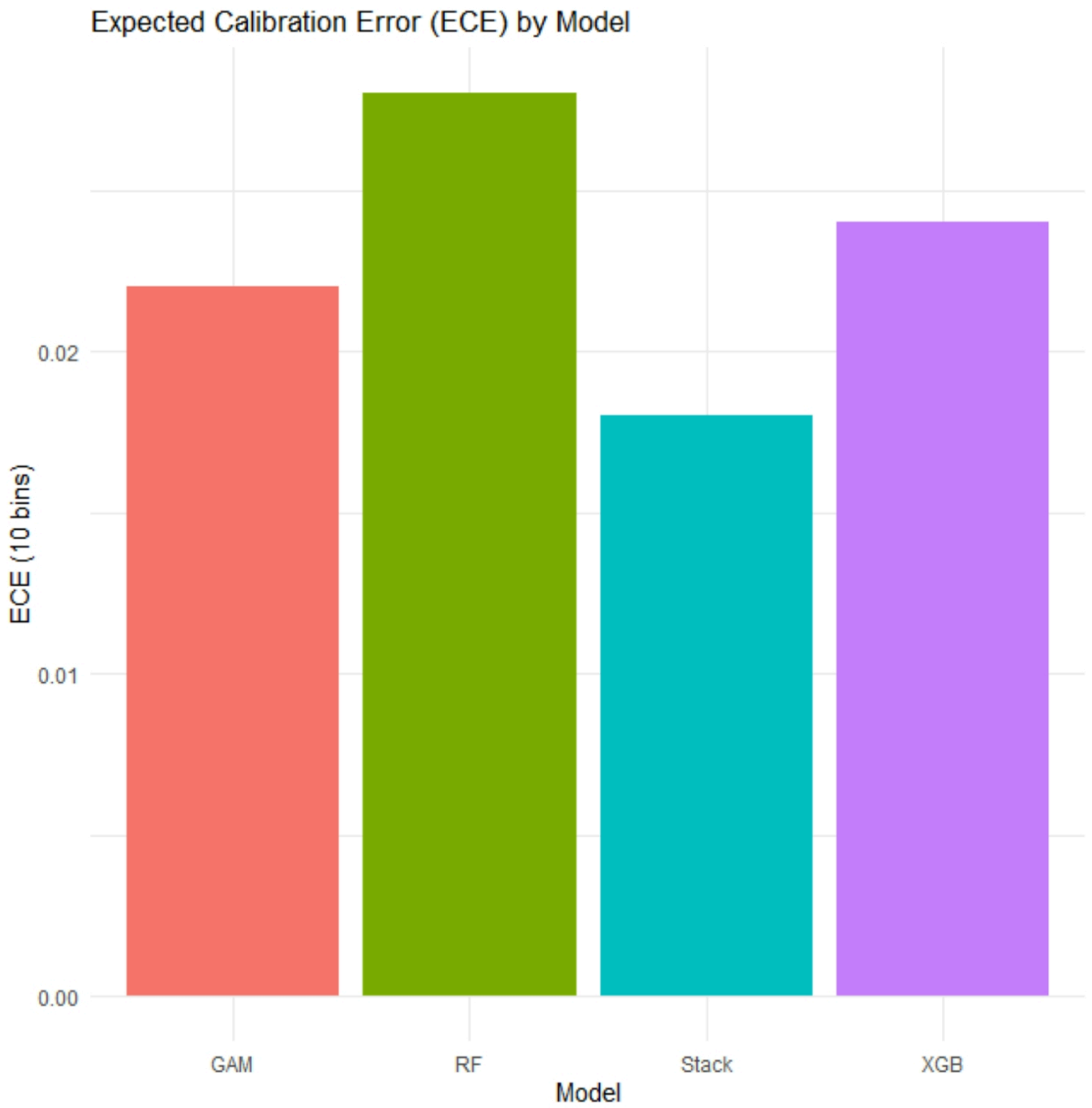


Figure 2

Model Calibration: Expected Calibration Error (ECE

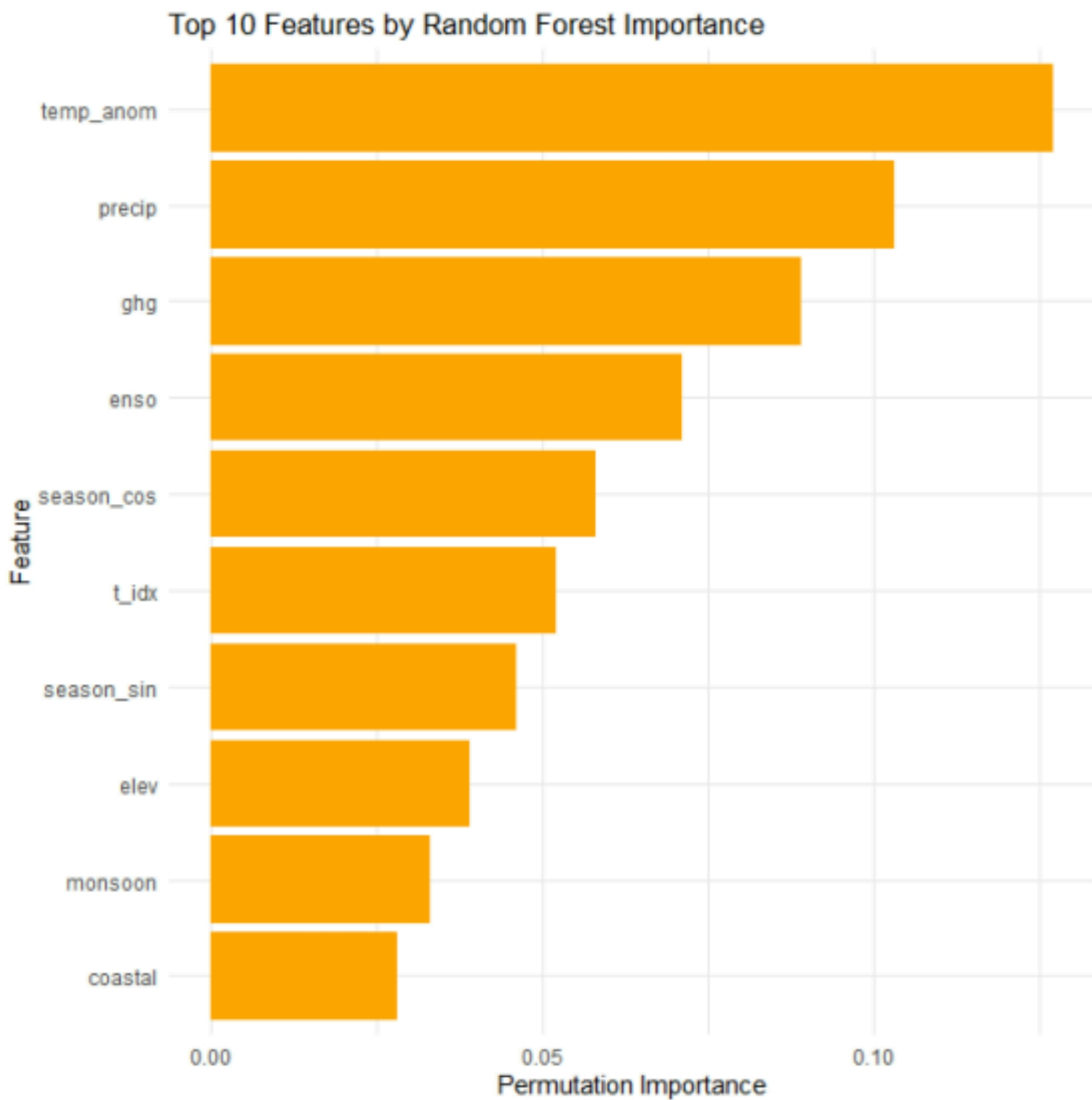


Figure 3

Top 10 Features by Random Forest Permutation Importance

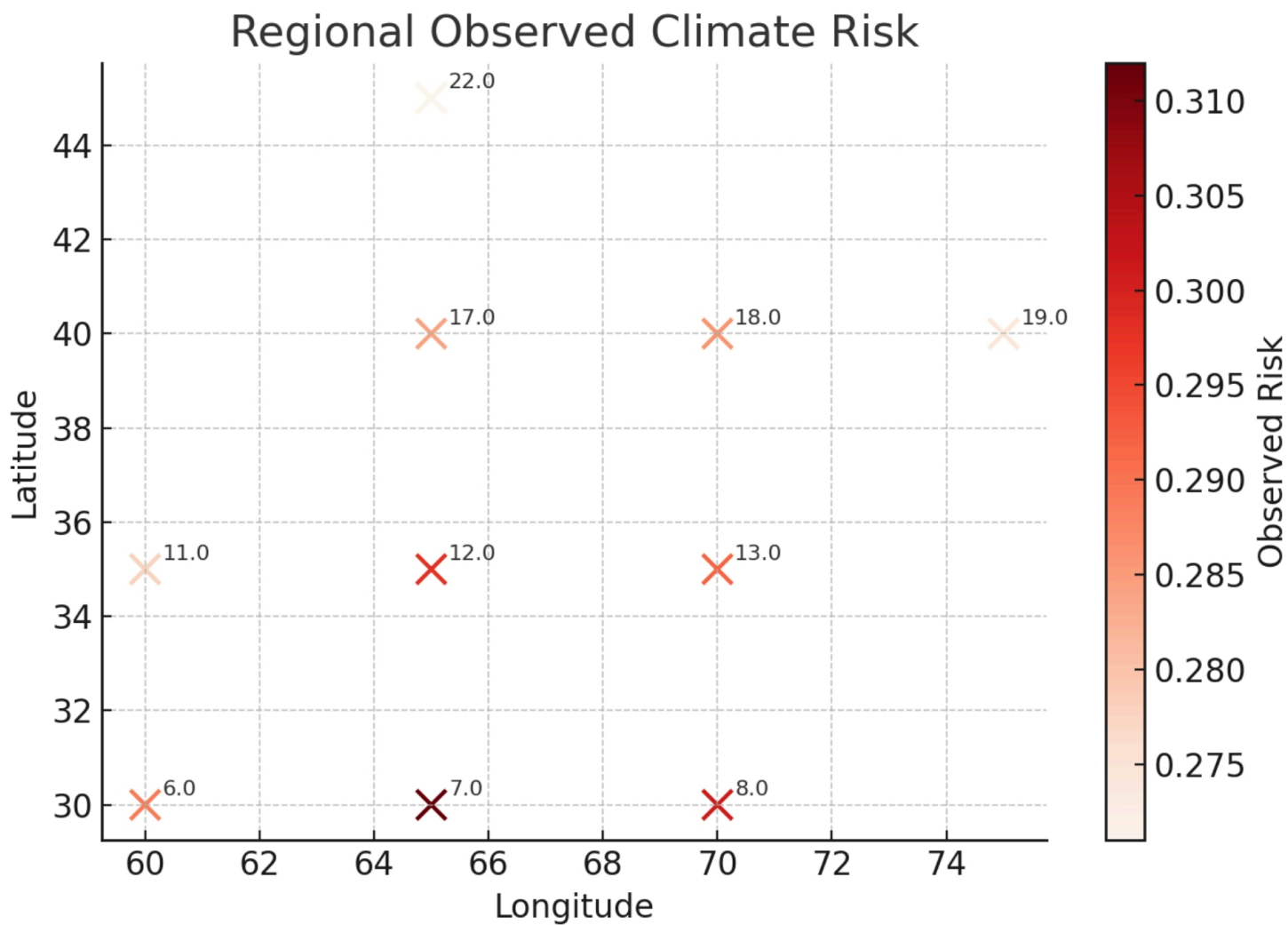


Figure 4

Regional Risk Map

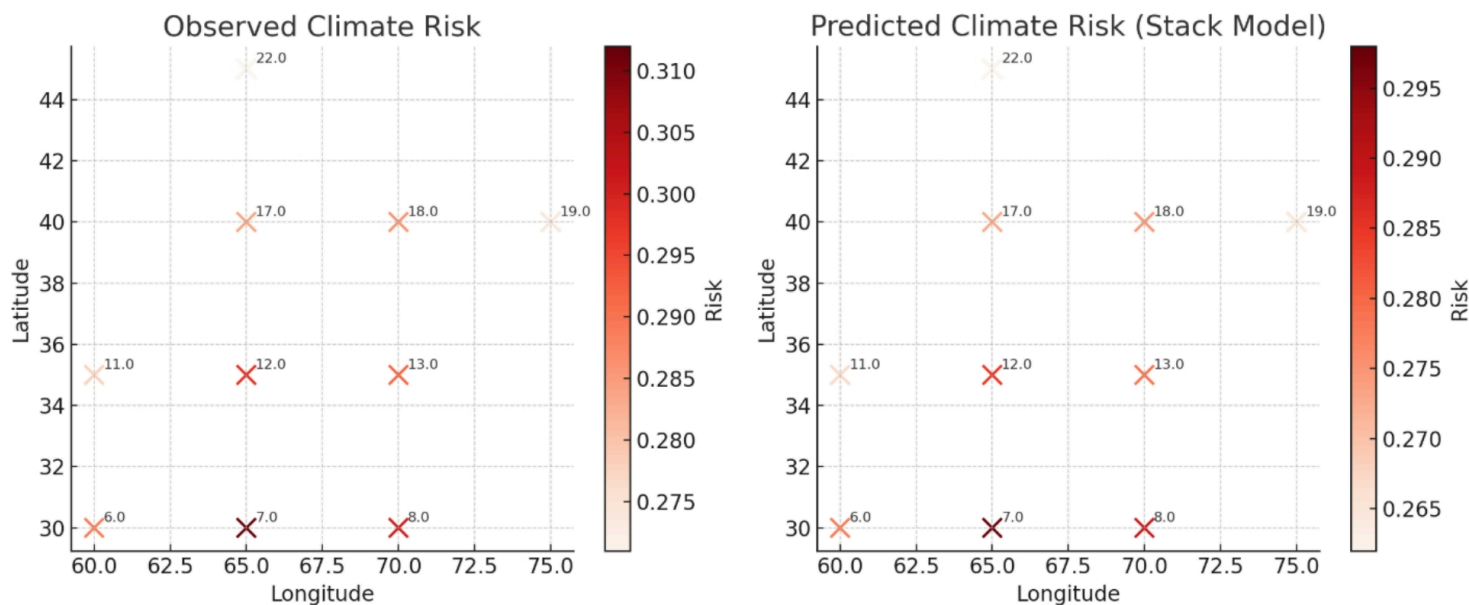


Figure 5

Side-by-Side Comparison of Observed Risk and Stack Model Predicted Risk

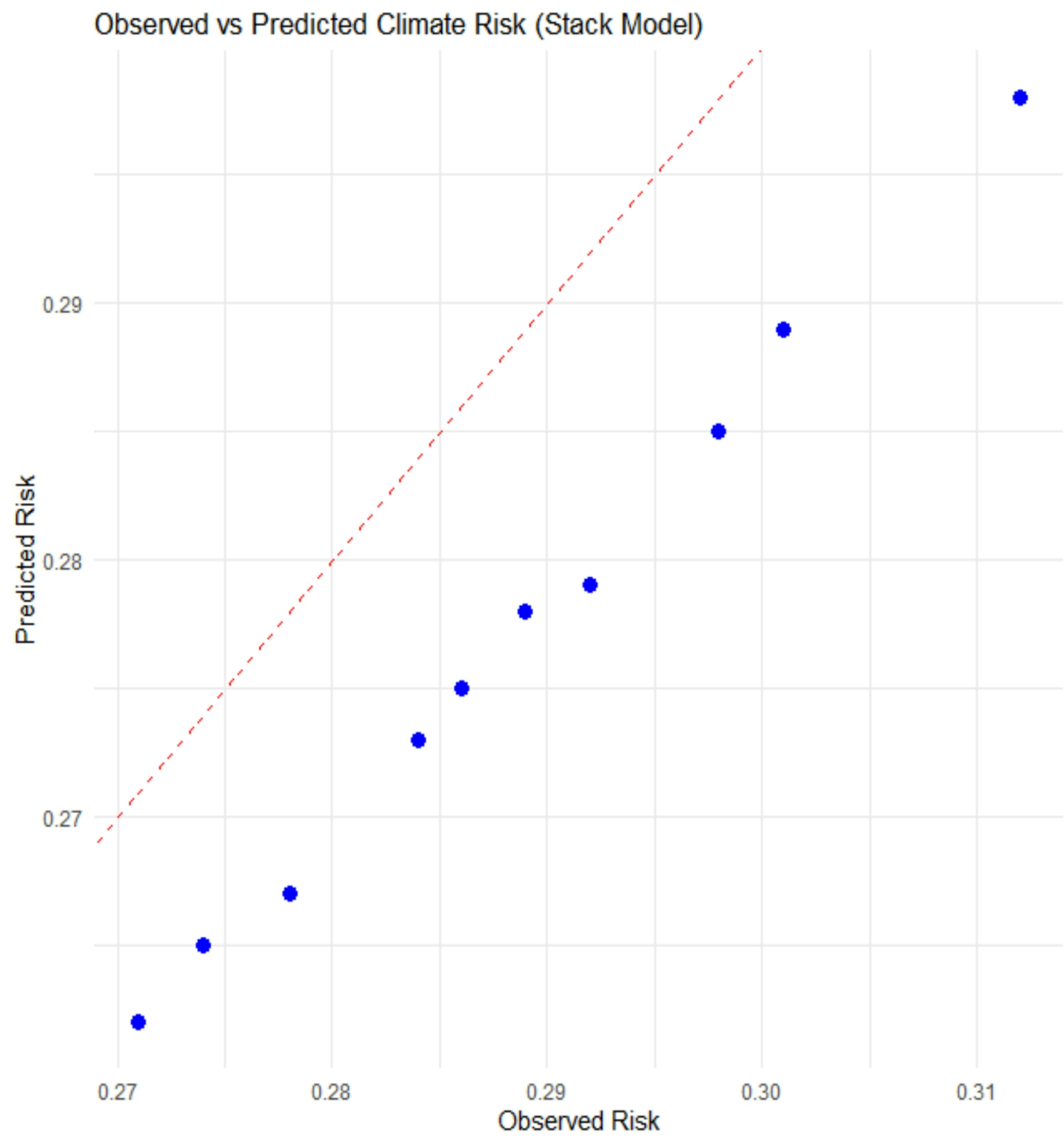


Figure 6

Observed vs. Predicted Climate Risk